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Chemical Engineering Journal

Date: 15 November 2025

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Polysulfide-mediated solid-state wetting for low-pressure all-solid-state lithium metal batteries

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Outline

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Highlights

- Polymerized LA exhibits binding affinities to Li metal, sulfide SE, and Li—Sn alloy.
- LA–Sn system enables thermodynamically favorable solid-state wetting of Li metal.
- Sn fragmentation drives depth-wise redistribution throughout bulk Li metal matrix.
- Reversible S—S bonds in LA provide mechanical resilience and Li₂S protective barrier.

- Stable operation at 2.5MPa and room temperature with 3.5 mAh cm^{-2} at 0.5C.

Abstract

The quest for practical all-solid-state lithium metal batteries (ASSLMBs) has focused on achieving reliable operation at room temperature and low stack pressure. However, attempts to lower this pressure have proved challenging due to interfacial instabilities and unwanted lithium (Li) dendritic growth. Here, we introduce a solid-state wetting approach of the Li metal anode by utilizing a polysulfide derived from α -lipoic acid and lithiophilic tin seeds. The polymeric network constructs strong interfacial bridges between the Li metal and solid electrolyte, while simultaneously driving the depth-wise redistribution of the pulverized Sn seeds throughout the bulk Li metal, thereby achieving comprehensive solid-state wetting that extends across the anode structure. This architecture, by concurrently mitigating the interfacial and bulk instabilities in $\text{Li}_{0.97}\text{Mg}_{0.03}$ metal anodes, enables stable operation at room temperature and low stack pressure (2.5MPa), along with exceptional capacity retention of 86.7% over 160 cycles at 0.5C with a high areal capacity of 3.5 mAh cm^{-2} . This study establishes foundational guidelines for the design of Li metal anodes toward high-performance ASSLMBs under practical operating conditions.

Graphical abstract

CRediT authorship contribution statement

Declaration of competing interest

Acknowledgements

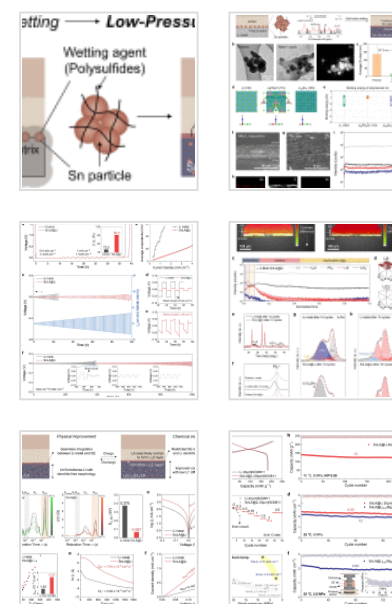
Appendix A. Supplementary data

Data availability

References

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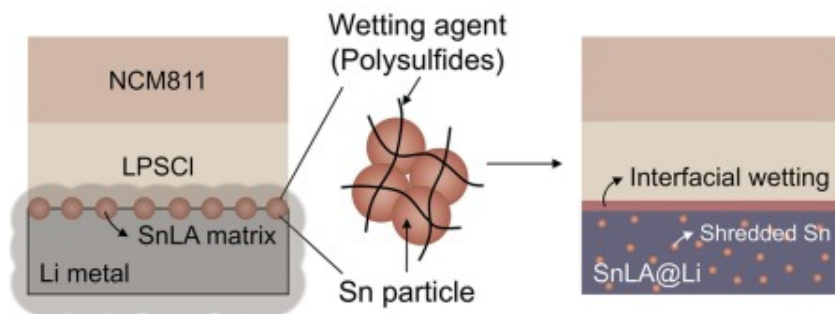
Figures (6)



Extras (1)

 Supplementary material

Solid-Solid Wetting → Low-Pressure Operation



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Keywords

All-solid-state batteries; Lithium metal anodes; Low stack pressure; Solid-state wetting; Polysulfides; Lithiophilic seeds

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1. Introduction

The global transition toward electrified transportation has intensified research interest in all-solid-state batteries (ASSBs) as next-generation energy storage systems owing to their inherent safety benefits and potential superior energy density [1,2]. These advanced systems, in which flammable liquid electrolytes are replaced with safer solid-state alternatives, circumvent the critical safety limitations of conventional lithium (Li)-ion batteries (LIBs), thereby eliminating the primary origin of spontaneous ignition and, consequently, thermal runaway [3,4]. Among the various solid electrolyte (SE) options, argyrodite-type sulfides stand out as suitable candidates owing to their decent ionic

conductivity, which is critical for delivering key performance specifications in cell operation [5,6]. Moreover, their intrinsic mechanical ductility facilitates intimate electrode–electrolyte contact and accommodates the substantial volume changes associated with high-capacity electrode materials [7], [8], [9].

In pursuit of high-energy-density batteries, Li metal is well-suited owing to its intrinsic electrochemical properties represented by its high theoretical specific capacity (3860 mAh g^{-1}) and lowest reduction potential (-3.04 V vs. the standard hydrogen electrode) [10,11]. However, the substantial Li storage capacity, in turn, leads to severe volume changes during cycling [12], [13], [14]. Critically, Li metal anodes devoid of a (typically carbon-based) host material—so-called hostless anodes—are inevitably plagued by inhomogeneous nucleation and the growth of metallic Li around localized electrochemically active sites during deposition, which leads to the uncontrolled formation of Li dendrites and eventual internal short-circuiting [15,16]. In addition, the high reactivity of Li induces undesirable side reactions with electrolytes [17,18]. All these challenges accelerate the accumulation of electrically isolated (“dead”) Li, which impairs the Coulombic efficiency (CE) in each cycle, and presents urgent issues that require compelling remedies [19].

Various effective solutions have been introduced to delay the degradation of metallic Li, particularly in the form of a protective interlayer and lithiophilic seeds, both of which promote uniform Li metal (de)plating. Conventionally, this interlayer is typically confined to the surface of the Li metal anode where it modulates the interfacial energy between the anode and SE, and enables the construction of a controlled solid-electrolyte interphase (SEI) [20]. The seeds dictate the reaction dynamics based on their solubility in Li and thus influence the Li plating behavior [21]. Among the various seeds, metals that are soluble in Li, such as silver (Ag) and magnesium (Mg), demonstrate solid-solution behavior upon reacting with Li to ensure the formation of a single phase as they dissolve within the Li metal [22], [23], [24], [25]. Alternatively, seeds of insoluble crystallogens (i.e., group 14 elements such as silicon) first form intermetallic alloys with Li, which subsequently facilitate even Li deposition through preferential Li-ion adsorption and

strong binding affinity [26,27]. However, addressing the problems associated with either the interface (via surface-localized treatments) or the bulk Li metal alone has proved insufficient to sustain the reversibility of Li metal (de)plating during cycling with its demands for elevated temperatures and high stack pressures as prerequisites, neither of which is commercially feasible.

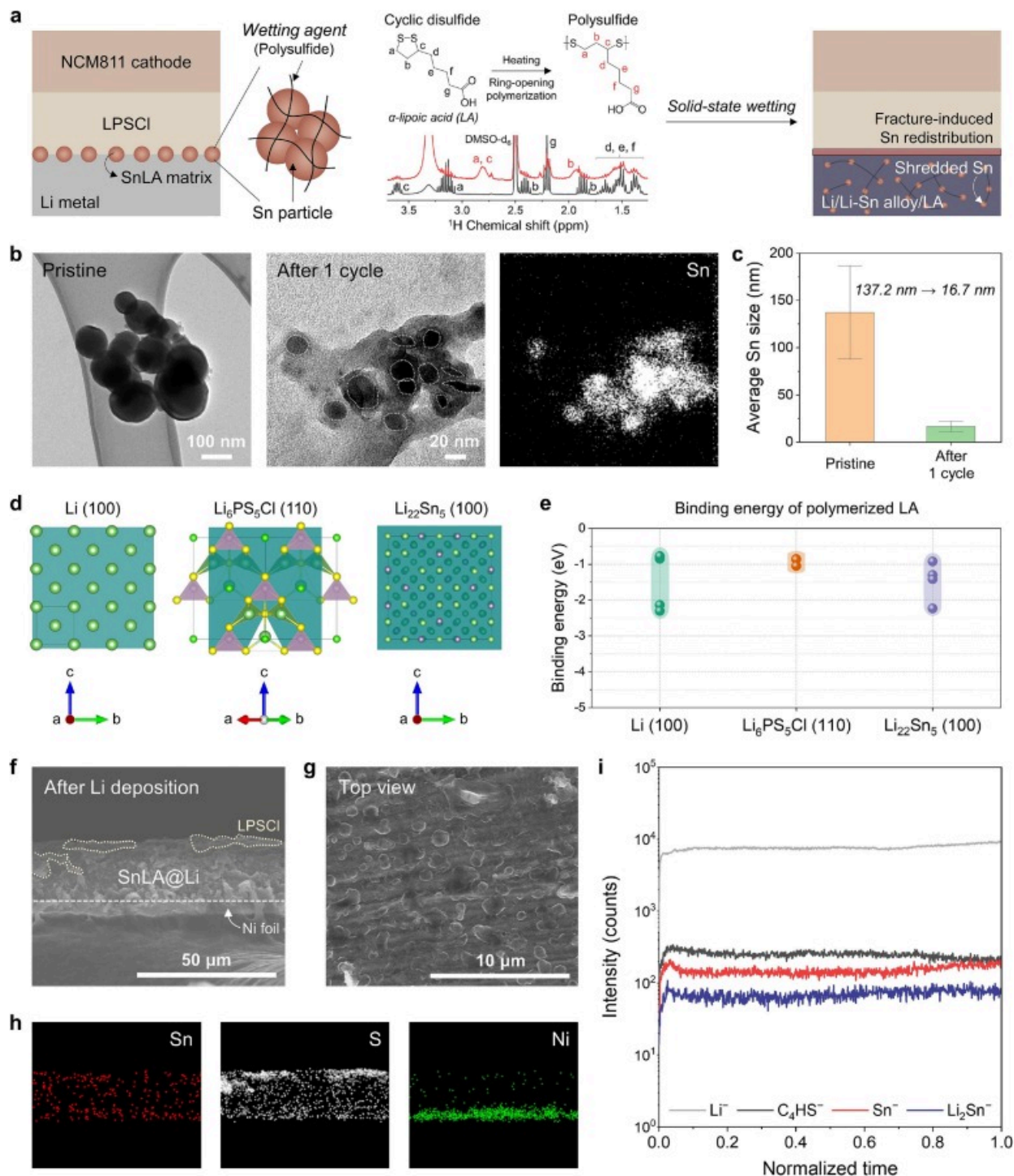
Herein, we propose a solid-solid wetting strategy that allows all-solid-state lithium metal batteries (ASSLMBs) to be operated at low stack pressure and room temperature. The strategy entails the use of a polysulfide derivative of α -lipoic acid (LA) to modify the bulk Li metal by seamlessly dispersing fractured lithiophilic tin (Sn) seeds in a depth-wise direction that extends from the surface to the core. In addition, the polymerized network promotes physical integration between the Li metal and the sulfide-based SE through polar interaction, thereby facilitating both interfacial and bulk wetting of the Li metal anode. Apart from this, the sulfur-containing polymeric LA selectively reacts with incoming Li at the superficial layer to preferentially form a thin, protective Li_2S film that chemically restricts SE decomposition and incipient Li dendritic growth. This structure ushers in operation at room temperature and low pressure (2.5 MPa) with a high loading capability (3.5 mAh cm^{-2}). Our approach lays the cornerstone for building advanced electrode systems by confronting long-standing challenges that have been hampering the development of high-performance ASSLMBs operable under practical conditions.

2. Results and discussion

2.1. Enabling the redistribution of fractured Sn seeds by solid-state wetting

Warranting the electrochemical reversibility of the Li metal anode in ASSLMBs is not trivial. This led us to fabricate a submicron-level SnLA framework by incorporating lithiophilic Sn seeds with polymeric LA (Fig. 1a and Fig. S1, S2). The cyclic disulfide of LA undergoes ring-opening polymerization via the dissociation of the S—S bond in the dithiolane ring following heat treatment at 70°C (Fig. S3). The polymerization creates a polysulfide network of a dynamic covalent nature with the ability to modify its

intramolecular structure through the reversible S—S bond of LA, thereby developing self-adaptive mechanical resilience under repeated Li plating/stripping [28]. The structural transformation, verified by ^1H nuclear magnetic resonance (NMR) spectroscopy, was evident from the substantial reduction of the cyclic resonances of $-\text{CH}_2-$ at 1.87, 2.40, and 3.13 ppm and the emergence of the characteristic peaks of the linear polymeric chain at 1.93 and 2.80 ppm [29]. The as-polymerized LA displays distinctive wetting characteristics with both Li metal and the sulfide SE, and serves as an interactive and dynamic matrix rather than a conventional inert interlayer. The phenomenon of solid-state wetting [30]—a fundamental concept in ASSB research—governs effective solid-solid contact at interfaces critical for cell operation, particularly under low stack pressure [31,32]. Here, the function of LA extends beyond that of a mere interfacial mediator by exhibiting both surface and bulk wetting capabilities. This comprehensive action not only establishes spontaneous adhesion at the SE|Li metal interface without reliance on an external mechanical force but also stabilizes the bulk Li metal by redistributing the Sn-based nucleation seeds.



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Fig. 1. Design principle of solid-state wetting for fracture-induced Sn redistribution. (a) Schematic illustration showing the 3D redistribution of the SnLA matrix, highlighting the inevitable fracturing of Sn particles and polysulfide-mediated Li metal wetting. (b) TEM images and the corresponding EELS mapping of Sn particles: pristine and after 1 cycle. (c) Size distribution analysis of Sn particles: pristine and after 1 cycle. (d) Atomic models of the low-energy facets of Li, LPSCI, and $\text{Li}_{22}\text{Sn}_5$ used for DFT binding energy calculations. (e) DFT-calculated binding energies of polymerized LA on Li(100), LPSCI(110), and $\text{Li}_{22}\text{Sn}_5$ (100) surfaces. (f) Cross-sectional and (g) top-view SEM images of the SnLA matrix after Li deposition, with (h) the corresponding EDS mapping and (i) ToF-SIMS depth profiling results.

In the first lithiation process, the lithiophilic Sn seed undergoes multiple crystallographic phase transitions, based on the Li alloying reaction to sequentially form distinct crystalline phases: $\text{Sn} \rightarrow \text{Li}_2\text{Sn}_5 \rightarrow \text{LiSn} \rightarrow \text{Li}_7\text{Sn}_3 \rightarrow \dots \rightarrow \text{Li}_{22}\text{Sn}_5$ [33,34]. This imparts a huge amount of mechanical stress to the particle that triggers inevitable pulverization even at domains as small as 10 nm, with theoretical modeling predicting critical sizes smaller than a single unit cell [35], [36], [37]]. After 1 cycle in a SnLA/SE/Li asymmetric cell configuration, Sn fragmentation becomes apparent from a comparison of the dimensions in the pristine state and after 1 cycle through TEM analysis, along with electron energy loss spectroscopy (EELS) mapping (Fig. 1b and Fig. S4). Specifically, the average size of a Sn particle decreased from 137.2 nm to 16.7 nm after 1 cycle, proving its mechanical vulnerability to electrochemical reactions (Fig. 1c). This mechanical pulverization leverages the unique compatibility of polysulfides to promote the conformal distribution of the resulting Sn fragments throughout the Li metal. This simultaneously ensures strong interfacial adhesion with the SE and optimizes the function of these fragments as lithiophilic seeds [38]. Furthermore, the scattered-island configuration of the SnLA matrix is beneficial for accommodating alloying-induced stress. The stress generated by the Li—Sn alloying reaction inevitably disrupts both Sn particles and LA networks. Therefore,

these discrete SnLA islands readily fragment and subsequently redistribute into Li/Li-Sn alloy/LA domains, facilitated by dynamic stabilization that enables the Li-ions to diffuse into the shredded matrix and accumulate in the interstices of the alloy (Fig. S5) [39]. Consequently, the even distribution of Sn seeds in the anode leads to homogeneous Li metal storage, and is attributed to its low Li-ion diffusion barrier and mixed ionic-electronic conducting properties [40,41].

Computational validation of the dual functionality of LA as both a wetting agent at the SE|Li metal interface and a promoter of multidirectional Sn seed redistribution involved systematic binding energy analyses using density functional theory (DFT) calculations. The selection of complementary surfaces—Li(100) and LPSCl(110)—was guided by their respective minimal surface energies (Fig. 1d). The structure of the $\text{Li}_{22}\text{Sn}_5$ alloy, which represents the fully lithiated phase observed in our electrochemical system, was employed to model Sn seeds embedded in the Li metal. Surface energy evaluations across the various crystallographic orientations identified the (100) surface of $\text{Li}_{22}\text{Sn}_5$ as the most thermodynamically stable configuration (Fig. S6), establishing it as the representative model. Calculation of the interfacial binding energy at various representative binding sites across different material junctions consistently yielded favorable values ($< -0.7\text{ eV}$), thus indicating robust adhesion across all tested configurations (Fig. 1e). Therefore, starting from the wetting of the SE|Li metal interface mediated by LA, bulk Li metal wetting proceeds via the relocation of fractured Sn seeds, promoted by thermodynamically favorable LA– $\text{Li}_{22}\text{Sn}_5$ coordination, the dynamical covalent adaptability ($-\text{S}-\text{S}-$) of the deformable LA, and its chemical affinity toward Li (Fig. S7). Indeed, the wettability of molten Li on the SnLA layer was significantly improved compared to that on a bare nickel (Ni) current collector (Fig. S8), confirming the solid-state wetting behavior of the SnLA layer with Li metal.

Based on these structural features, the SnLA accommodates incoming Li, demonstrating its ability to guide compact Li maturation and yield a smooth morphology devoid of protrusions, even for substantial Li storage (Fig. 1f, g and Fig. S9). Cross-sectional SEM images with energy dispersive spectroscopy (EDS) mapping analysis confirmed an even

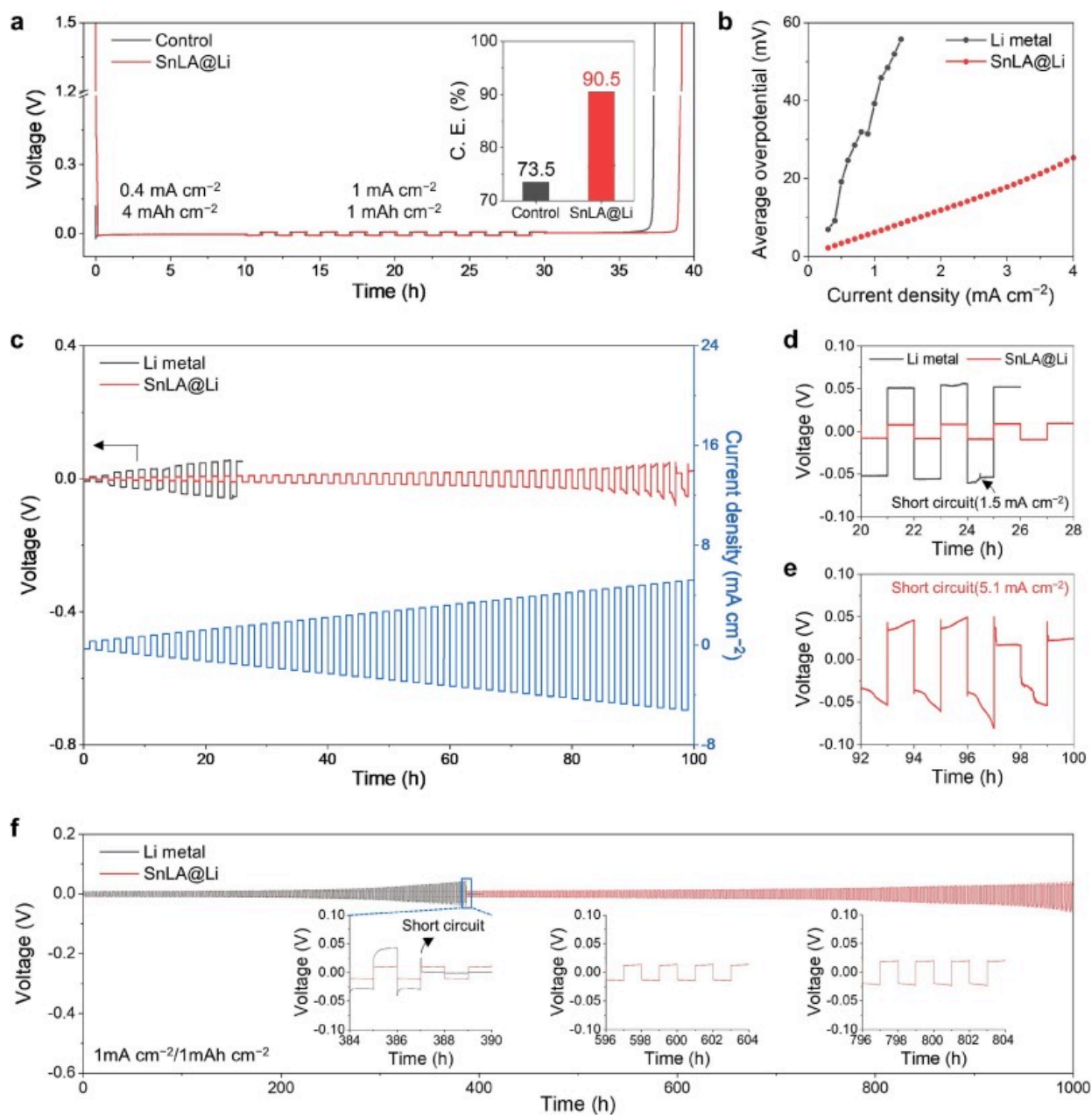
dispersion of Sn fragments throughout the depth of the Li deposition layer (Fig. 1h).

Unlike Sn seeds, Si—a prototypical crystallogen seed—undergoes simplified crystallographic phase changes upon Li insertion by inherently resisting electrochemical fragmentation below ~150nm and consequently limiting its spatial redistribution for bulk Li wetting. This behavior results in the restricted dispersion of Si within the Li metal matrix, as observed for SiLA@Li (Fig. S10). In contrast, Ag—a canonical lithiophilic seed—readily forms solid solutions with Li via a fundamentally different electrochemical mechanism [42,43]. Although this mechanism enables the uniform distribution of Ag in AgLA@Li, it does not offer the intermetallic stabilization characteristic of crystallogen seeds. Examination in a Li/SnLA/SE/SnLA/Li symmetric cell configuration confirmed these fragments to be uniformly distributed throughout the Li metal matrix after activation cycles (Fig. S11). Notably, in the absence of the LA wetting agent, the lopsided and heterogeneous distribution, rather than the seamless dispersion, of the Sn matrix was attributed to limited Li deposition between the Li—Sn alloy domains (Fig. S12). The spatial distribution of chemical species was investigated through time-of-flight secondary ion mass spectrometry (ToF-SIMS) depth profiling (Fig. 1i). The presence of C_4HS^- from LA, alongside Sn^- and Li_2Sn^- bonds indicative of the Li—Sn alloy, was consistently observed throughout the depth, revealing that the bulk Li metal had been wetted by islands of SnLA.

2.2. Assessment of electrochemical behavior

The growth of dendritic Li and low CE characteristic of Li metal anodes are primarily responsible for cell malfunction. Therefore, the reversibility of the Li metal plating/stripping process in SnLA@Li was assessed using the Aurbach protocol (Fig. 2a) [44,45]. The CEs of the anode with SnLA@Li and the control sample without any interfacial treatment were conspicuously distinct at 90.5% and 73.5%, respectively. To decouple the contributions, we evaluated Sn@Li and LA@Li separately. Sn@Li reached 85.5% through Li—Sn alloy-mediated stabilization, whereas LA@Li increased the CE to 91.5% via LA-derived Li_2S passivation at the Li|SE interface (Fig. S13). Collectively, these

results indicate that the SnLA matrix fosters dense, uniform Li deposition while confining the parasitic side reactions that consume the Li sources.



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Fig. 2. Electrochemical evaluation of SnLA@Li. (a) Li metal plating/stripping CEs for the control and SnLA@Li using the Aurbach protocol. (b) Relationship between current density and average overpotential. (c) Voltage profiles under stepwise current for CCD measurements of Li metal and SnLA@Li symmetric cells (Stack pressure=8 MPa). Magnified profiles showing short circuit behavior for (d) Li metal and (e) SnLA@Li. (f) Voltage profiles of symmetric cells with Li metal and SnLA@Li anodes (current density: 1 mA cm^{-2} ; capacity: 1 mAh cm^{-2}).

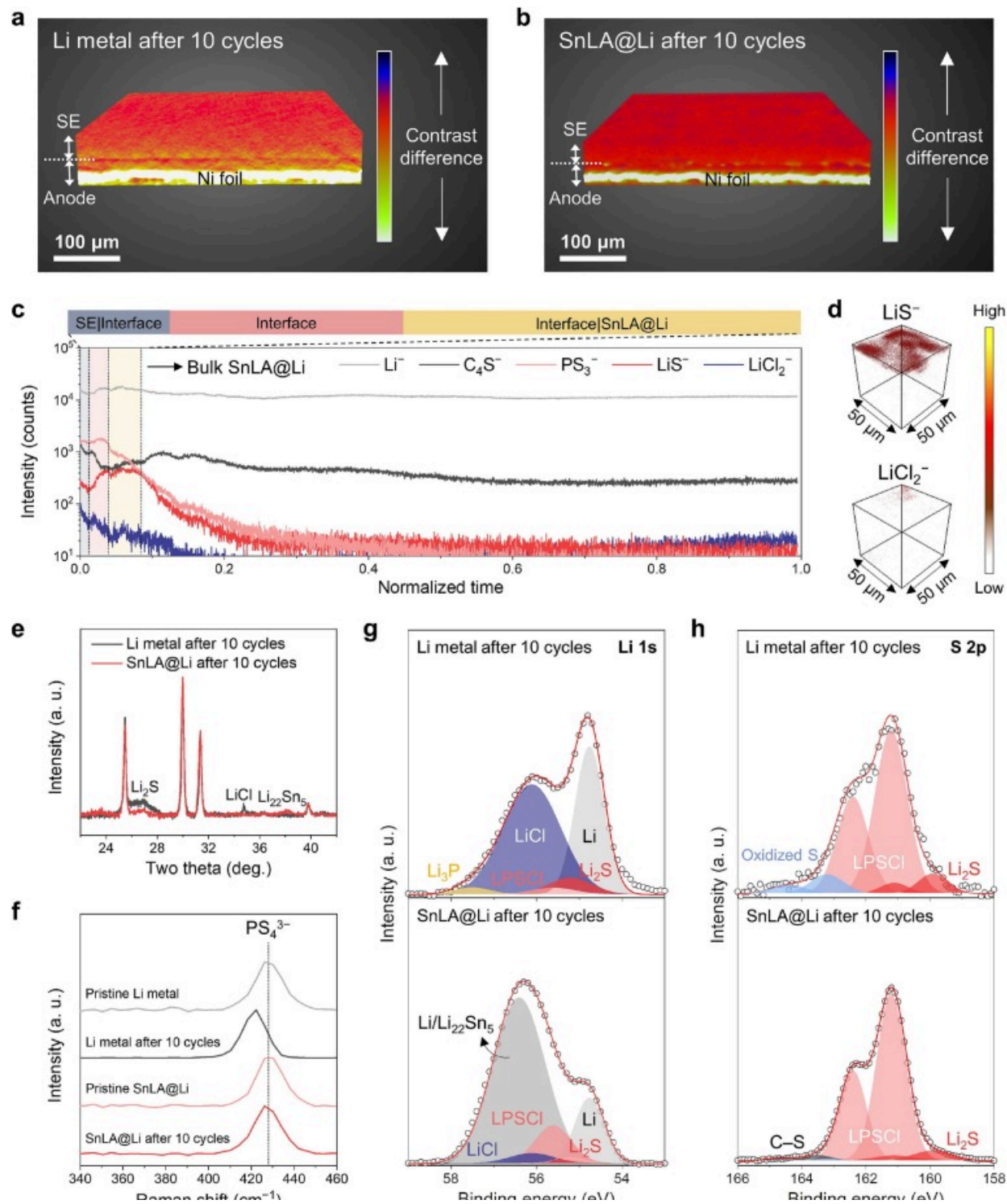
The ability of SnLA@Li to withstand high current densities was further evaluated by increasing the current density stepwise to determine the critical current density (CCD) (Fig. 2b, c). First, various Sn/LA mass ratios (3:1, 5:1, 7:1, and 9:1) were systematically evaluated, with the 9:1 ratio identified as optimal due to the synergistic balance between Sn and LA (Fig. S14). The overpotential of the SnLA@Li (9:1) cell was significantly lower than that of the bare Li metal anode at equivalent current densities. This behavior suggests the efficient modulation of Li-ions and Li metal at elevated current densities to reach a notably higher CCD of 5.1 mA cm^{-2} (Fig. 2d, e). In contrast, the Li metal cell short-circuited at 1.5 mA cm^{-2} , indicating that pristine Li metal is susceptible to failure even when operated at moderate rates. Additionally, Sn@Li and LA@Li were examined under identical conditions to elucidate the contribution of each component for the purpose of comparison (Fig. S15). The CCD values for Sn@Li and LA@Li reached 4.0 and 3.2 mA cm^{-2} , thereby revealing the effectiveness of their individual constituents. Evidently, by employing lithiophilic crystallogen seeds, the formation of an intermetallic Li—Sn alloy inside the Sn@Li matrix provides efficient pathways for Li-ion transport while reinforcing the interfacial stability. Furthermore, the affinity of LA for the solid-state components gives rise to effective wetting, which imparts outstanding resilience under extreme current conditions. This strategic integration of the wetting medium and dispersible lithiophilic seeds from SnLA@Li enables stable operation while mitigating the risk of short circuits. Similarly, CCD comparisons with SiLA@Li and AgLA@Li yielded the trend: SnLA@Li (5.1 mA cm^{-2}) > SiLA@Li (3.7 mA cm^{-2}) > AgLA@Li (2.0 mA cm^{-2}) (Fig. S16). This

hierarchy can be rationalized by the different behaviors of the seeds: Ag dissolves into Li to form weak-binding solid solutions that preserve Li mobility but lead to structural instability under high current densities. In contrast, crystallogenic seeds such as Si and Sn create strong-binding intermetallic alloys with Li, stabilizing the SE|anode interface [20,27]. Specifically, SnLA@Li—having achieved depth-wise wetting—not only retains its role as a dynamic interfacial layer but also serves as a Li metal host that establishes preferential adsorption sites across the anode, steering bulk densification and thus enhancing electrochemical accessibility. The cycling stability afforded by the SnLA framework was evaluated at constant current density and constant capacity of 1 mA cm^{-2} and 1 mAh cm^{-2} , respectively, in symmetric cell configurations (Fig. 2f). The Li plating/stripping overpotential of the bare Li metal cell gradually increased due to inadequate interfacial contact and continuous decomposition of the SE, which led to dendrite-induced short-circuiting in less than 400h. In contrast, the SnLA@Li anode displayed outstanding longevity of more than 1000h. The remarkable cycle life of this anode highlights the multifunctionality of the SnLA architecture, in particular its ability to ensure homogeneous Li accumulation and the formation of a robust interface.

2.3. Polysulfide-mediated interfacial wetting

X-ray microscopy (XRM) combined with micro-computed tomography was used for 3D visualization of the SE|anode interfaces (Fig. 3a, b). The distinctive interfacial features were validated by contrast mapping based on the X-ray transmission levels. The bare Li metal electrode exhibited discontinuous contrast variations at the SE|anode interface, whereas the SnLA@Li electrode demonstrated a consistent contrast pattern, indicative of a monolithic structure throughout the system. Importantly, the formation of the monolithic structure is attributed to the coordination of Li ions in $\text{Li}_6\text{PS}_5\text{Cl}$ (LPSCl) with the polar sulfur groups in the polysulfides of LA [38,46]. Hence, seamless integration was established through the solid-solid wetting at the electrode–electrolyte boundary. In addition, localized contrast variations observed within the SE layer of the bare Li metal anodes signified the formation of electrochemically inactive “dead” Li or secondary SEI phases—both indicative of dendritic infiltration [24]. In contrast, cells incorporating the

SnLA protective layer exhibited homogeneous SE layer contrast, demonstrating the suppression of dendritic penetration.



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Fig. 3. Interfacial analysis of SnLA@Li for sustainable operation. XRM images of SE|anode interfaces of (a) Li metal and (b) SnLA@Li after 10cycles. (c) ToF-SIMS depth profiling results of SnLA@Li after Li deposition and (d) 3D mapping of LiS^- and LiCl_2^- . (e) XRD patterns of SE|anode interfaces for Li metal and SnLA@Li after 10cycles. (f) Raman spectra of SE|anode interfaces for Li metal and SnLA@Li: pristine and after 10cycles. XPS (g) Li 1s and (h) S 2p spectra of SE|anode interfaces for Li metal and SnLA@Li after 10cycles.

The distribution of chemical species across distinct regions was analyzed using ToF-SIMS depth profiling (Fig. 3c, d). The low concentration of LiCl_2^- , as evidenced by the low-intensity peak detected at the SE|interface, gradually subsided toward the bulk SnLA@Li region. Note that the initial decomposition of the LPSCI typically begins with the formation of LiCl as the onset of systematic deterioration [47,48]. However, significant amounts of LiCl were not observed throughout the structure. Moreover, the consistent presence of PS_3^- , extending from the SE|interface to the interface with the SnLA@Li region, did not noticeably increase, indicating that SE decomposition reactions were being restrained. Importantly, the inverse correlation between the increasing LiS^- (signifying Li_2S generation) and decreasing C_4S^- (characteristic of LA) concentrations at the interface, together with the constant PS_3^- and LiCl_2^- signals, confirmed that Li_2S was derived from LA rather than originating from the reduction of the SE. The Li_2S pre-formed at the anode interface by the SnLA layer constitutes a thin interphase—on the order of tens of nanometers—that passivates the Li|SE contact and thereby suppresses the degradation of the sulfide-based SE upon contact with Li metal. In the absence of detrimental reactions, the parallel behavior of C_4S^- and PS_3^- at the SE|interface demonstrates the chemical compatibility between LA and LPSCI, while their favorable polar interactions stabilize the interfacial region.

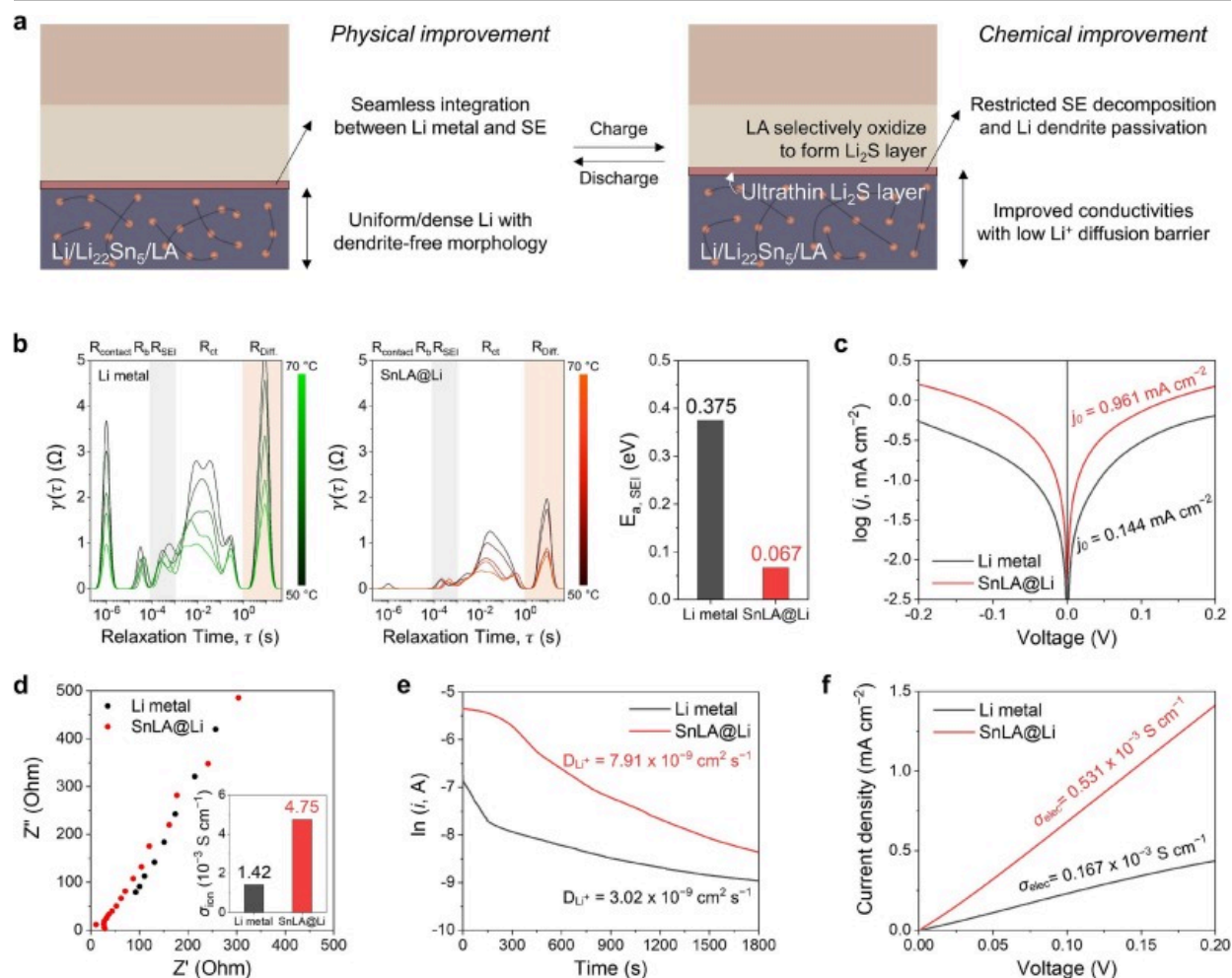
Concurrently, the XRD patterns confirmed the emergence of $\text{Li}_{22}\text{Sn}_5$ (38.2°) and Li_2S (26.9°) from SnLA@Li , without any LiCl signals ($34.7\text{--}34.8^\circ$) (Fig. 3e) [40,49,50]. In the case of bare Li metal, both the LiCl and Li_2S phases were detected as undesired byproducts. The Raman spectra further unveiled that the bare Li electrode drastically degraded the SE, evidenced by a shift of the PS_4^{3-} peak to lower values (from 428cm^{-1} to 421.5cm^{-1}) after 10cycles, whereas SnLA@Li maintained the consistent spectral features of PS_4^{3-} (Fig. 3f) [50]. Post-cycling Li 1s XPS analysis revealed that bare Li metal predominantly gave rise to Li_2S (55.2eV) and LiCl (56.1eV) with the formation of Li_3P (57.5eV) and trace amounts of LPSCl (55.6eV) (Fig. 3g) [[51], [52], [53], [54]]. In contrast, the introduction of SnLA@Li had the effect of retaining the characteristics of the SE, with the presence of minimal LiCl alongside Li metal (54.8eV) and $\text{Li/Li}_{22}\text{Sn}_5$ as the dominant Li-containing component (56.4eV) (Fig. 3h) [40]. The S 2p spectra also corroborated the formation of Li_2S (159.9 and 161.1 eV) in SnLA@Li from the polysulfide (C—S , 163.5 and 164.7 eV), where the resulting superficial layer preserves the integrity of the SE (161.2 and 162.4 eV) and prevents the development of resistive interfacial phases such as oxidized S (i.e., $\text{P—[S]}_n\text{—P}$ and PS_3O^{3-} ; 163.2 and 164.4 eV), with P 2p analysis showing consistent stabilization (Fig. S17) [55,56]. Additional Raman (Fig. S18), XPS (Fig. S19), and SEM–EDS (Fig. S20) analyses of SnLA@Li after 50cycles showed the same trends as those observed in early cycles, confirming the persistent structural integrity and interfacial stability.

The interfacial characteristics of Sn@Li and LA@Li were compared to elucidate the contributions of the individual constituents to the electrochemical reactions (Fig. S21). The non-uniform distribution of Sn in Sn@Li resulted in the localized formation of $\text{Li}_{22}\text{Sn}_5$ —as corroborated by the cross-sectional SEM–EDS images in Fig. S12—which, in the case of Sn@Li , does not serve as a host or matrix for Li storage (i.e., $\text{Li/Li}_{22}\text{Sn}_5$) but rather functions as a simple interfacial layer, thereby enabling the undesirable accumulation of pure Li metal beneath it [20]. Similarly, XPS depth profiling after 10cycles revealed that $\text{Li/Li}_{22}\text{Sn}_5$ was scarcely distributed within the interior of Sn@Li , whereas, in the case of SnLA@Li , a uniform and substantial presence of $\text{Li/Li}_{22}\text{Sn}_5$ developed across the entire depth from the surface inward (Fig. S22). Nevertheless, Sn@Li limited the unwanted

reactivity with the SE by forming $\text{Li}_{22}\text{Sn}_5$ in the bulk structure, irrespective of its heterogeneous aggregation (Fig. S23). Without the incorporation of Sn, LA@Li achieved solid-solid wetting and protected its interface by preferentially covering Li_2S to mitigate the decomposition of the SE.

2.4. Physicochemical origins for electrode kinetics

The unique physicochemical properties of SnLA@Li boost its electrochemical function as a Li reservoir, as illustrated in Fig. 4a. From a physical perspective, the dynamic evolution at the anode/SE boundary stems from the coordination chemistry between SE and the polysulfides, which allows them to integrate seamlessly. The optimized Sn seeds contribute to preventing spatial mismatches by enabling a uniform and dense accumulation of Li metal with a dendrite-free morphology. Additionally, the strategic combination extends to the chemistry of SnLA@Li, where an LA-derived ultrathin Li_2S layer acts as a thermodynamic barrier between the anode and SE. Here, the SnLA@Li serves to confine the breathing surface, which is prone to detrimental self-discharge. Notably, the evenly distributed $\text{Li}_{22}\text{Sn}_5$ in SnLA@Li supplements both the ionic and electronic conductivity while improving the tolerance against SE decomposition to ensure consistent Li storage capability.



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Fig. 4. Electrochemical characterization of SnLA@Li reflective of the physicochemical advancements. (a) Schematic of the interfacial and bulk benefits of SnLA@Li in ASSLMs. (b) Temperature-dependent DRT transition curves from EIS measurements of Li metal and SnLA@Li, with corresponding bar graphs displaying the activation energy barriers derived from the Arrhenius fitting of R_{SEI} . (c) Tafel plots for Li metal and SnLA@Li, along with the corresponding exchange current densities. (d) Impedance measurements of Li metal and

SnLA@Li with the electron-blocking cells (Inset: comparison of calculated ionic conductivity). (e) Current evolution plots for PITT tests and calculated Li-ion diffusion coefficients. (f) LSV curves for Li metal and SnLA@Li, with calculated electronic conductivity.

The impact of the SnLA matrix was demonstrated by temperature-dependent electrochemical impedance spectroscopy (EIS) analysis and the corresponding distribution of relaxation time (DRT) transition (Fig. 4b and Fig. S24) [[57], [58], [59]]. The contact resistance (R_{contact}) of SnLA@Li at the relaxation time ($\tau=10^{-6}$) is nearly eliminated owing to the sulfur-assisted interfacial wetting of LA chains that anchor the SE at the boundary region [60,61]. This was further supported by the shredded and dispersed crystallogen Sn seeds, contrary to the poor contact that is chronically maintained by the bare Li metal, as indicated by the pronounced red-to-white color gradient. Moreover, the high bulk resistance (R_{bulk} , $\tau=10^{-5}$ – 10^{-4}) and SEI resistance (R_{SEI} , $\tau=10^{-4}$ – 10^{-3}) attest to severe side reactions between the SE and bare Li metal, which negatively impact the cycling performance and lower the CE. In contrast, for SnLA@Li, both the bulk and interfacial resistance decreased markedly owing to the enhanced solid-solid wetting. Likewise, the Arrhenius plot revealed that bare Li metal had a high activation energy barrier (E_a) of 0.375 eV for Li transport through the SEI region, whereas the significantly lower value of 0.067 eV for SnLA@Li highlighted its kinetic superiority. The considerably distinct charge transfer resistance (R_{ct}) values of SnLA@Li at $\tau=10^{-3}$ – 10^0 imply that its combined ionic/electronic conducting properties are electrochemically advantageous in terms of the redox kinetics. Intriguingly, the mass diffusion resistance ($R_{\text{diffusion}}$) of SnLA@Li decreased markedly at $\tau=10^1$, which can be linked to 3D Li maturation beneath the surface around the Sn seeds in the interior, as opposed to Li plating on the top surface of bare Li metal. These trends were consistent with those revealed by low-temperature (15–35 °C) EIS–DRT analysis (Fig. S25).

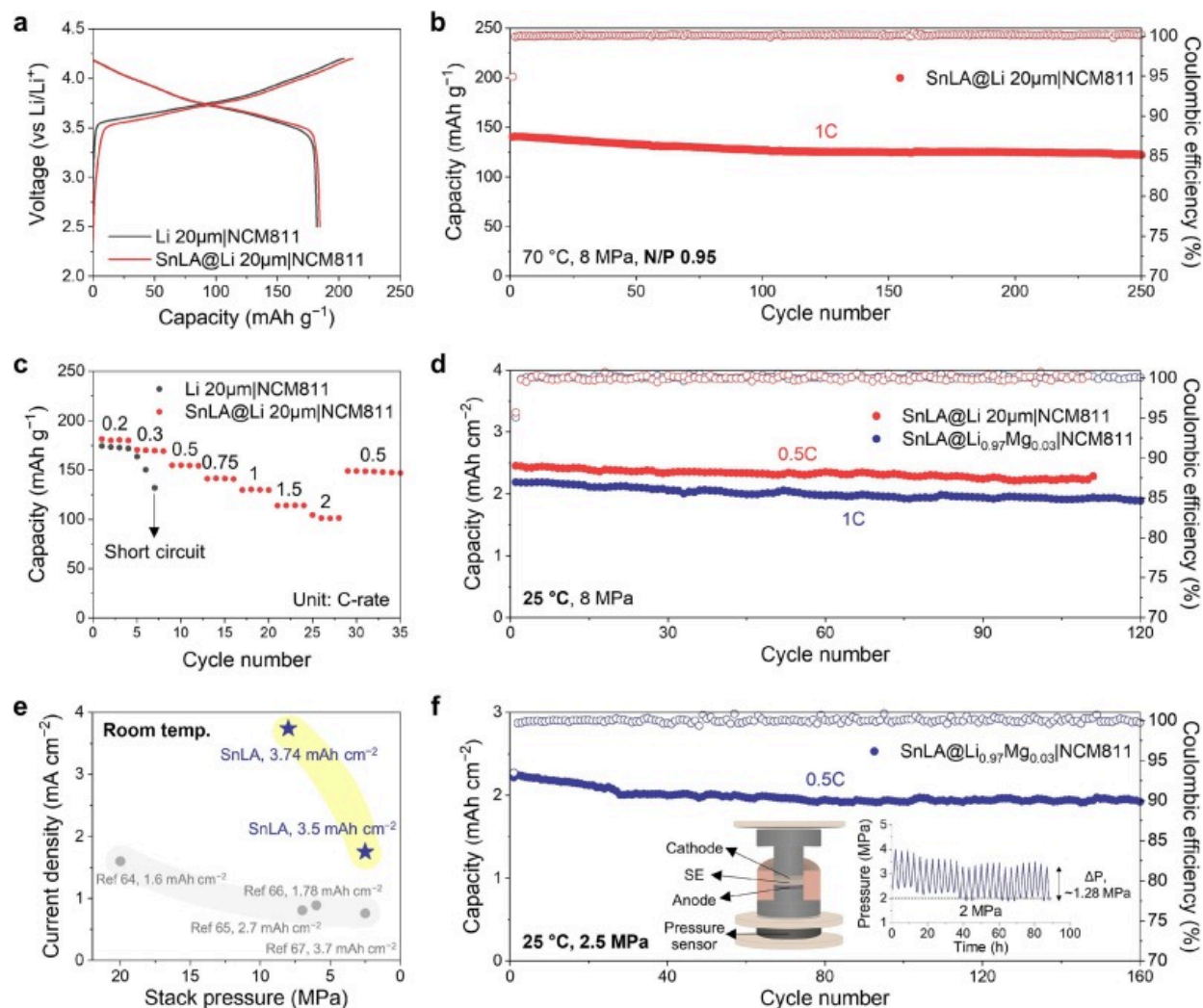
The interplay between the impedance and electrochemical features was investigated by comprehensively analyzing the kinetic properties. To this end, the Tafel curves for symmetric cells were recorded at –5 °C, as acquiring them at room temperature would

have generated a strong current that would have caused the cell to short circuit (Fig. 4c) [62]. The bare Li metal produced a weak exchange current of 0.144 mA cm^{-2} , whereas that of SnLA@Li was substantially amplified to 0.961 mA cm^{-2} . This enhancement is ascribed to LA-mediated interfacial integration and the formation of the Li—Sn alloy, both of which facilitate charge transfer at the interface as well as in the bulk Li metal matrix. The electrochemical behavior was additionally analyzed by deploying an electron-blocking cell structure to evaluate the resistance to ionic movement (Fig. 4d) [63]. The ionic conductivity of SnLA@Li ($4.75 \times 10^{-3} \text{ S cm}^{-1}$) was surprisingly high relative to that of the bare Li metal ($1.42 \times 10^{-3} \text{ S cm}^{-1}$). Similarly, the Li-ion diffusion coefficient, calculated using the potentiostatic intermittent titration technique (PITT), reached $7.91 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1}$ for SnLA@Li, in contrast to $3.02 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1}$ for the bare Li electrode (Fig. 4e). The extent to which the electronic properties of SnLA@Li benefited from these improved kinetics was explored by utilizing linear sweep voltammetry (LSV) (Fig. 4f). The measurement revealed that the current density in SnLA@Li was augmented during Li deposition. Specifically, the electronic conductivity of SnLA@Li ($0.531 \times 10^{-3} \text{ S cm}^{-1}$), which surpassed that of the bare electrode ($0.167 \times 10^{-3} \text{ S cm}^{-1}$), can be explained by polysulfide-mediated solid-state wetting and the resulting bulk-distributed $\text{Li}_{22}\text{Sn}_5$ that serve as mixed ionic/electronic conductors. This physicochemical regulation of the Li behavior is responsible for increasing the CCD and cycling durability.

2.5. Practical ASSLMB operation

Given the reinforced interfacial stability and electrochemical characteristics of the SnLA matrix, diverse full-cell configurations based thereupon were assessed to confirm their applicability for practical ASSLMB systems. The SnLA layer was integrated with a thin layer of Li metal ($20 \mu\text{m}$) to serve as the anode and paired with an NCM811 cathode (areal capacity: $\sim 4 \text{ mAh cm}^{-2}$) at an N/P ratio of 0.95 under mild stack pressure (8 MPa). The beneficial effect of solid-state wetting was evident from the EIS measurements (Fig. S26), which revealed substantially lower internal resistance in the SnLA-containing cell compared to that of its bare Li metal-based counterpart before and after the rest period, consistent with the LA–SE interaction results reported in Fig. 3a and Fig. 4b. In terms of

reversible capacity, the Li 20 μ m|NCM811 and SnLA@Li 20 μ m|NCM811 full cells initially delivered comparable performance of 182.2 and 184.5 mAh g⁻¹ at 0.1C (Fig. 5a). At an elevated rate of 1C, the SnLA@Li 20 μ m|NCM811 displayed exceptional longevity by retaining 86.9% of its capacity over 250cycles (Fig. 5b). Despite its fast-charging capability, decreasing the Sn-to-LA mass ratio to 2:1 led to cell failure within 18cycles at 1C, validating that the optimal ratio was 9:1 (Fig. S27). Under the modest pressure of 8MPa, the Li 20 μ m|NCM811 cell experienced internal short-circuiting during early cycling, thus indicating that operation at 1C was prohibitive (Fig. S28). Based on the Nyquist plots derived from the EIS analysis, the internal resistance of the two cells differed considerably after a selected number of cycles (Fig. S29). The SnLA@Li 20 μ m|NCM811 cell demonstrated sustainable anode integrity at 1C even under harsh cycling conditions represented by a high areal capacity of 4 mAh cm⁻² and a low N/P ratio of 0.95. In contrast, the resistance of the Li 20 μ m|NCM811 cell significantly increased during the early cycling period because of soft shorts. The rate capability measurements further revealed the superiority of SnLA@Li 20 μ m|NCM811 (N/P 0.95), which retained 72.4% and 56.3% of its capacity at 1C and 2C relative to that at 0.2C, respectively (Fig. 5c). However, the failure of the Li 20 μ m|NCM811 cell at merely 0.3C suggested insufficient endurance for high-rate operation. Under a moderate cathode loading (2mAhcm⁻²), the SnLA@Li 20 μ m|NCM811 also exhibited long-term cycling stability at 2C with capacity retention of 70.4% after 800cycles (Fig. S30).



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Fig. 5. Electrochemical performance of full cells with the SnLA matrix, operated under practical conditions. (a) Galvanostatic charge/discharge profiles of the Li 20 μ m|NCM811 and SnLA@Li 20 μ m|NCM811 full cells at 0.1C under a low N/P ratio system, with the (b) corresponding cycling stability at 1C and (c) rate capability at various rates for both charge and discharge. (d) Room temperature cycling stability of the SnLA@Li

20 μ m|NCM811 full cell at 0.5C and SnLA@Li_{0.97}Mg_{0.03}|NCM811 full cell at 1C. (e)

Comparison of stack pressure, current density, and areal capacity of full cells against those of previously reported ASSLMs at room temperature. (f) Room temperature cycling stability under low stack pressure of the SnLA@Li_{0.97}Mg_{0.03}|NCM811 full cell at 0.5C (Inset: schematic of the pressure sensing system and corresponding in situ pressiometry result during cycling).

At room temperature, the SnLA@Li 20 μ m|NCM811 cell demonstrated a prolonged cycle life at 0.5C by retaining 90.8% of its capacity after 100cycles (Fig. 5d). In contrast, the Li 20 μ m|NCM811 cell experienced short-circuiting during the second cycle at 0.5C, implying that operation at ambient temperature would be unsustainable (Fig. S31). Collectively, these results show that the robustness of the SnLA matrix extends the performance of the corresponding cell beyond the prerequisites for operating ASSLMs at elevated temperatures, and they also prove its versatility. Leveraging the known solid-solution behavior between Li and Mg, the incorporation of Li_{0.97}Mg_{0.03} foil into the SnLA matrix was expected to enhance the reversibility of Li metal (de)plating at room temperature. In fact, SnLA@Li_{0.97}Mg_{0.03} enabled stable operation at an increased rate of 1C with capacity retention of 86.3% over 120cycles (Fig. 5d). Notably, without the SnLA layer, the capacity of the Li_{0.97}Mg_{0.03}|NCM811 cell decayed rapidly to below 80% within three cycles at 1C and an internal short circuit occurred after 13cycles (Fig. S32). This highlights the critical function of the SnLA matrix with its ability to realize reliable reversibility at high current densities.

Establishing sustainable solid-solid contact in ASSLMs under low stack pressure during cycling is nontrivial due to their substantial interfacial degradation as a result of the mechanical instability of the interfaces. Nevertheless, the reversibility of SnLA@Li 20 μ m|NCM811 proved to be robust under commercially viable conditions (room temperature and 2.5MPa; Fig. S33). Additionally, to further improve the cycling performance, a Li—Mg composite anode with 3%Mg doping (Li_{0.97}Mg_{0.03}) was employed. As a result, despite employing the high-loading NCM811 cathode (3.5 mAh cm⁻²), the cycling stability of the SnLA@Li_{0.97}Mg_{0.03}|NCM811 configuration was outstanding even at

0.5C with capacity retention of 86.7% over 160 cycles (Fig. 5e, f). Specifically, in situ pressiometry measurements enabled the operational behavior to be visualized at the lowest stack pressure of 2.5 MPa. The refined design of the SnLA matrix effectively overcomes the chronic issue of severe void formation and the accompanying rise in the resistance by exhibiting stable pressure resilience (ΔP) of ~ 1.28 MPa. In the absence of the SnLA framework, the $\text{Li}_{0.97}\text{Mg}_{0.03}\text{|NCM811}$ cell immediately short-circuited during the initial cycles (Fig. S34). When benchmarked against previously reported ASSLMs, our SnLA-based system demonstrates unprecedented performance metrics under practically feasible operating parameters, particularly in terms of the stack pressure, current density, and areal capacity at room temperature [[64], [65], [66], [67]].

3. Conclusions

In summary, we demonstrated ASSLMs capable of exceptional performance with a high areal capacity combined with operability at room temperature and low stack pressure (2.5 MPa). Our approach exploits the strong interfacial binding of the polysulfide LA network, which cohesively integrates the Li metal anode and sulfide SE. Importantly, the structural adaptability of the LA framework, coupled with its molecular-level interaction with Li—Sn alloys, facilitates the dynamic redistribution of fragmented Sn seeds throughout the bulk Li metal to realize the depth-wise wetting of the Li metal anode. Concurrently, the polymeric LA matrix forms a protective Li_2S barrier through selective surface reactions to mitigate electrolyte decomposition and suppress dendritic growth. This dual stabilization at both the interface and in the bulk anode addresses critical bottlenecks in ASSLMs by enabling them to deliver high cycling stability under practical operating conditions. This study presents solid-state wetting of Li metal anodes as a transformative concept for achieving sustainable ASSLMs for next-generation energy storage systems.

CRedit authorship contribution statement

Minjun Je: Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis. **Jihun Sung:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis. **Jihoon Oh:** Writing – original draft, Validation, Investigation, Formal analysis. **Heejin Kim:** Investigation, Formal analysis. **Dongmin Park:** Validation, Investigation. **Hyoyeong Kim:** Validation. **Youngjin Song:** Validation. **Sang-Min Lee:** Investigation. **Jang Wook Choi:** Writing – original draft, Supervision, Project administration, Funding acquisition. **Soojin Park:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Materials and Components Technology Development Program of MOTIE/KEIT (Grant No. [20007045](#)) and the Korea Innovation Foundation funded by Ministry of Science and ICT ([RS-2023-IN231032](#)). J.W.C. acknowledges the National Research Foundation of Korea (NRF) (Grant No. [RS-2023-00261543](#)) and the Swiss National Science Foundation (SNF) (Grant No. Sinergia [CRSII5_202296](#)). J.W.C. also acknowledges support from the Institute of Engineering Research (IOER), the Research Institute of Advanced Materials (RIAM), and the Institute for Battery Research Innovation (IBRI) at Seoul National University.

Appendix A. Supplementary data

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Supplementary material

Data availability

Data will be made available on request.

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